RTC 2027: Electronics MDR Checklist

Student 3K Category

Avionics • Recovery Electronics • Official GPS • Additional Mission Payload • Electrical Integration

MDR deadline: 13 November 2026. This is an internal checklist for the electronics team's contribution to the **single team MDR**, not a request to submit separate subsystem reports. The overall MDR is limited to **21 pages** and is submitted as a searchable PDF. **At MDR:** propose and justify the design and verification plan; completed flight-ready hardware and final verification evidence belong to later reviews.

Source key (files from the uploaded RTC package)

Code	Filename	Document
A01	`appendix_01.pptx`	Mission Plan Report example / slide template
A03	`appendix_03.pdf`	Rocket Design Specifications
A06	`appendix_06.pdf`	Competition Rules Explanation, version 2
A07	`appendix_07.xlsx`	MDR scoring and Design Maturity Matrix
A08	`appendix_08.xlsx`	Launch Permit Inspection Checklist
A10	`appendix_10.xlsx`	Requirements Matrix Example
A12	`appendix_12.xlsx`	FMEA Example
A13	`appendix_13.xlsx`	Rocket Design Specification Data Sheet
A14	`appendix_14.xlsx`	Rocket Sensor Item List

How to use: Tick each box once its content is prepared for the MDR. Citations identify the underlying competition rule, example template or scoring criterion. Items labelled "propose", "describe", or "plan" are MDR work products; they do not assert that hardware testing is already complete.

1 Mission objectives and functional requirements

Related MDR scoring: mission objectives and system requirements (20 marks each).

Done	MDR checklist item	Source document and location
☐	Define what the rocket electronics must accomplish during the 3K mission.	A01, Slides 2 and 9; A07, worksheet `03_MDR`
☐	Define measurable success, partial-success and failure criteria for avionics and the additional mission payload.	A07, worksheet `03_MDR`, Row 6
☐	List functional requirements for avionics, recovery electronics, official GPS and the additional payload.	A01, Slide 9; A07, worksheet `03_MDR`, Row 7
☐	Link each functional requirement to a mission objective and a responsible subsystem.	A07, worksheet `03_MDR`, Row 7; A10
☐	Define the four required flight-data outputs: altitude, velocity, acceleration and GPS trajectory.	A03, Section 2.1
☐	Create a mission profile showing flight phases, altitude, time, recovery events and payload operations.	A01, Slides 9–10; A06, PDF page 18

2 Avionics system

Flight computer, sensing, flight-state decisions and logging.

Done	MDR checklist item	Source document and location
☐	Prepare a block diagram showing the flight computer, sensors, storage, power and external interfaces.	A07, worksheet `03_MDR`, Row 8; A01, Slide 9
☐	Propose a controller, barometer, IMU and onboard data storage, with a brief selection rationale.	A06, PDF pages 21 and 23; A07, worksheet `03_MDR`
☐	Explain how launch, ascent, apogee and descent will be detected.	A03, Section 4.2.2.4; A01, Slide 11
☐	Describe how altitude, velocity, acceleration and GPS trajectory will be obtained and recorded.	A03, Section 2.1
☐	Propose sampling rates, sensor ranges, logging formats, measurement units and timestamps. These are team-defined targets.	A07, worksheet Design Maturity Matrix (Sheet 14), Row 31; worksheet 3K Flight (Sheet 11), Row 19
☐	Explain how vibration, impact and sensor measurement error will be addressed.	A03, Section 4.1.2
☐	Describe how normal avionics operation and system/arming status will be confirmed.	A13, Rows 12–14; A03, Section 4.6.7

3 Recovery electronics and deployment

Two-stage deployment, activation reliability and safety.

Done	MDR checklist item	Source document and location
☐	Show the two-stage concept: drogue at apogee and main parachute at a selected lower altitude.	A03, Section 4.2.2.4
☐	Select a preliminary main deployment altitude, normally 150–450 m above ground, consistent with the below-500 m inspection criterion.	A03, Section 4.2.2.4; A08, Row 8
☐	Explain the proposed recovery-release mechanism and its electrical-control interface.	A03, Sections 4.2.1–4.2.2
☐	Show how independent or redundant recovery activation will be achieved.	A03, Section 4.1.3
☐	Show redundancy in the recovery release arrangement.	A03, Section 4.2.2.2
☐	Describe safe and armed states, independent safety conditions and safe arming-status confirmation.	A03, Sections 4.6.3–4.6.7
☐	Account for the final water-contact descent-speed limit of less than 12 m/s.	A03, Section 4.2.4; A08, Row 8
☐	Plan sensor-trigger, recovery-activation and representative environmental tests.	A03, Sections 4.2.2.3 and 4.5.1–4.5.3

4 Official GPS tracking and communications

The organiser's tracking beacon is distinct from the additional mission payload.

Done	MDR checklist item	Source document and location
☐	Identify the organiser-standard GPS tracker as a required 3K subsystem separate from the additional mission payload.	A03, Sections 2.2 and 5.1.1
☐	Show proposed GPS beacon location, mounting and electrical/physical interfaces.	A03, Section 5.1.1; A01, Slide 8
☐	Describe how GPS tracking supports rocket location and recovery after landing.	A03, Section 5.1; A08, Row 26
☐	Plan how normal GPS operation and position reporting will be demonstrated.	A03, Section 5.1.2; A08, Row 26
☐	If using extra radio telemetry, document the proposed frequency, messages, receiver and regulatory compliance.	A03, Sections 4.1.4.1–4.1.4.3

5 Additional mission payload and electronics

A team-defined payload with measurable flight success criteria.

Done	MDR checklist item	Source document and location
☐	Define the additional payload mission objective and its core function.	A03, Section 2.2; A07, worksheet `03_MDR`, Row 6
☐	Define measurable success, partial success and failure criteria for the payload.	A07, worksheet `03_MDR`, Row 6; worksheet 3K Flight (Sheet 11), Row 20; A13, Row 36
☐	Identify preliminary payload sensors, controller, storage, other hardware and operating sequence, if applicable.	A06, PDF page 23; A07, worksheet Design Maturity Matrix (Sheet 14), Row 15
☐	Choose the permitted operation configuration: independent operation with retention until parachute opening, or in-body operation with avionics communication throughout flight.	A03, Section 7.1.1
☐	Define payload electrical, data/communication and mechanical interfaces.	A06, PDF pages 21 and 23; A07, worksheet Design Maturity Matrix (Sheet 14), Row 15
☐	Explain compliance with the payload restriction on prohibited stored-energy devices.	A03, Section 7.1.2
☐	Specify payload battery requirements and applicable radio compliance where wireless transmission is used.	A03, Sections 7.1.3–7.1.4
☐	Confirm the additional payload does not contain living vertebrates.	A03, Section 7.1.5
☐	Explain what recorded measurements or other evidence will demonstrate mission completion in flight.	A07, worksheet 3K Flight (Sheet 11), Row 20

6 Power, wiring and electronic integration

Electrical endurance, mechanical support and subsystem interfaces.

Done	MDR checklist item	Source document and location
☐	Select a proposed battery chemistry and summarize the safety considerations.	A03, Section 4.7.1
☐	Prepare a power budget showing more than 1.5 hours of normal electrical operation.	A03, Section 4.7.2; A08, Row 6
☐	Describe battery mounting that prevents intermittent disconnection under acceleration or vibration.	A03, Section 4.7.3
☐	Specify secure safety-critical wiring, cable labels, connector strain relief and PCB support.	A03, Sections 4.4.2–4.4.5
☐	Identify subsystem power/data interfaces and any shared supplies or dependencies.	A06, PDF page 21; A07, worksheet Design Maturity Matrix (Sheet 14), Row 12
☐	Show the preliminary electronics location and estimated subsystem mass in the rocket concept.	A01, Slide 8; A07, worksheet Design Maturity Matrix (Sheet 14), Row 16

7 Budget, planning, risks and verification

Related MDR scoring: team/resources and initial risks/verification (20 marks each).

Done	MDR checklist item	Source document and location
☐	Prepare a preliminary electronics BOM with specifications, quantities, make-or-buy decisions and prices.	A01, Slides 6–7; A07, worksheet `03_MDR`, Row 9
☐	Assign owners for avionics, recovery electronics, GPS integration and payload electronics.	A01, Slides 3–4
☐	Prepare a Gantt chart for design, procurement, fabrication, programming, testing and integration.	A01, Slide 5; A06, PDF page 17
☐	Create a development flowchart showing subsystem dependencies and integration sequence.	A01, Slide 11
☐	Prepare a requirements-verification matrix with ID, source, owner, verification method and planned evidence.	A06, PDF page 24; A10
☐	Specify which requirements will be checked by inspection, demonstration, analysis or test.	A06, PDF page 24
☐	Develop a preliminary FMEA covering power loss, sensor error, processor failure, recovery failure, radio failure and payload malfunction.	A01, Slide 12; A12
☐	State key design uncertainties and how the team will resolve them.	A07, worksheet `03_MDR`, Rows 8 and 10

8 MDR versus later review stages

Summarized from A07, worksheet Design Maturity Matrix (Sheet 14); the distinctions below are planning guidance.

Design element	Prepare for MDR	Complete in later stages
Electronics architecture	Preliminary proposal	Design baseline at PDR
Avionics and payload specifications	Initial concepts and estimates	Finalized at CDR
Payload success criteria	Initial measurable baseline	Finalization and subsequent verification
Verification methods	Proposed methods	Supporting test results at PDR/CDR
Functional test evidence	Not required as completed MDR evidence	Verification at later reviews
FMEA	Preliminary risks and mitigations	Detailed analysis and verified mitigations
Wiring and PCB	Preliminary architecture/interfaces	Detailed design and hardware verification
Flight data	Not applicable at MDR	Post-flight reporting

9 Recommended electronics-team MDR handover

Suggested internal deliverables to integrate into the team's main report; not five separate mandatory TASA submissions.

1. **Electronics architecture:** one integrated diagram for avionics, recovery, GPS, additional payload, and their interfaces.
2. **Functional requirements matrix:** concise and measurable requirements with source and planned verification.
3. **Preliminary design and budget:** component selections, mass/power estimates, costed BOM, and design rationale.
4. **Development and verification roadmap:** responsibilities, milestones, integration sequence and planned tests.
5. **Initial electronics FMEA:** important failure modes, consequences and proposed mitigations.

MDR packaging reminder: Electronics content is included within the overall slide-style MDR submission. Refer to A01 for the presentation example, and A07 (worksheet Documents and Presentation (Sheet 02)) for the searchable-PDF format and 21-page cap. The proposal should show how the final flight hardware *will* meet the competition rules, without implying that later verification has already been completed.